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The impact of medication reconciliation on discrepancies and all-cause readmission among hospitalised patients with chronic kidney disease: a quasi-experimental study

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ABSTRACT

Background: Chronic kidney disease (CKD) with its associated comorbidities and pill burden can expose patients to a heightened risk of drug-related problems, including medication discrepancies. This study aimed to evaluate the impact of medication reconciliation supplemented with medication review on the number of medication discrepancies at discharge and all-cause readmission among CKD patients.

Methods: This was a quasi-experimental trial among adult CKD patients admitted into two major referral hospitals in northern Jordan. Patients in the intervention group received medication reconciliation supplemented with medication review by a clinical pharmacist, while those in the control group received the usual care. The recognised discrepancies were evaluated at admission and at discharge in both groups. Participants were followed for 90-day readmission.

Results: Among patients in the intervention group, the average number of discrepancies was 2.5 ± 2.2 per CKD patient. Compared to the control group, the reduction in discrepancy numbers between admission and discharge was higher in the intervention group by 1.66 discrepancies. The likelihood of 90-day readmission was significantly lower in the intervention group (OR = 0.41; $P = 0.002$).

Conclusion: Supplemented medication reconciliation among CKD patients reveals a favourable impact on medication discrepancies and readmission rates. Optimising medication management during transitions of care can improve overall health outcomes.

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Background

Medication reconciliation is a healthcare process in which healthcare providers partner with patients and/or their families/caregivers to ensure that the most accurate and complete medication list is maintained (*Canadian Patient Safety Institute (CPSI) and the Institute for Safe Medications Practices Canada (ISMP Canada), Medication Reconciliation in Acute Care Getting Started Kit edition 4*, 2017, 69P). This service aims to identify and resolve any drug discrepancies throughout the transitions of care in order to improve medication management (*Canadian Patient Safety Institute (CPSI) and the Institute for Safe Medications Practices Canada (ISMP Canada), Medication Reconciliation in Acute Care Getting Started Kit edition 4*, 2017, 69P). Medication discrepancies are classified as part of drug related problems (DRPs) and are defined as unexplained alterations between medication lists at transitions of care (*World Health Organization, Medication safety in transitions of care: technical report*, 2019). Medication discrepancies are among the primary causes of patient harm, potentially leading to adverse drug events (ADEs) and increased use of healthcare resources (Abu Farha et al., 2021; Chan et al., 2015; Quelenec et al., 2013; *World Health Organization, Medication safety in transitions of care: technical report*, 2019). Medication reconciliation combined with medication review offers additional protection against DRPs (Hammad et al., 2017), and leads to greater improvements in both clinical and economic outcomes (Najafzadeh et al., 2016).

In managing chronic diseases, medication reconciliation is crucial for optimising treatment outcomes and reducing complications linked to polypharmacy (Shrestha et al., 2019). It is a key component of patient safety and quality care, particularly for those with chronic conditions. By systematically reviewing and updating medication regimens, healthcare providers can minimise errors, enhance adherence, and improve overall health outcomes (Tariq et al., 2025). Collaborative care models in the healthcare system for chronic conditions enhance efficiency by ensuring smooth communication among providers during care transitions and fostering a multidisciplinary approach that involves physicians, pharmacists, and caregivers to improve patient care (Geese & Schmitt, 2023).

Patients with chronic kidney disease (CKD) are at increased risk of medication discrepancies and other DRPs since they are commonly elderly, have multiple comorbidities, require polypharmacy for multi-illness management, receive medications with high-alert risk, have problems with medication adherence, and are prone to frequent care transitions (Tesfaye et al.,

2019a). Interestingly, medication discrepancies have been demonstrated as a modifiable risk for CKD complications and progression (Sakhiya et al., 2020). Proper management of CKD necessitates collaborative efforts of a multidisciplinary healthcare team for optimising the medication review process (Kelepouris et al., 2023). Previous studies showed the critical role of clinical pharmacist services in achieving positive impact on health outcomes in CKD setting (Al Raiisi et al., 2019; Alshogran et al., 2022). However, data on the impact of medication reconciliation on discrepancies and related clinical outcomes remain limited, especially for inpatients with CKD (Al Raiisi et al., 2019; Whittaker et al., 2018), and warrant further investigations.

In developing countries, including Jordan, the evidence on effective CKD care is insufficient, and there is no clear consensus on the impact of practice models that involve pharmacists in the care of these patients. In Jordan, an average of 5.31 DRPs per patient among CKD inpatients (AbuRuz et al., 2013) and 6.92 DRPs per patient among haemodialysis patients (Shqeirat et al., 2021) were observed. However, discrepancies were not investigated in these local studies. This underscores the importance of assessing the impact of specialised clinical services on optimising CKD management and ultimately alleviating the significant clinical burden of this complex disorder. This study examines the role of clinical pharmacist in preventing harm caused by discrepancies, particularly in patients with CKD. This study aimed to evaluate the impact of medication reconciliation supplemented with medication review on the number of discrepancies at discharge and the likelihood of all-cause 90-day readmission in CKD admitted patients.

Methods

Study design and settings

This was a multi-center quasi-experimental clinical study involving patients with stages 2–5 CKD, and was conducted in two major referral hospitals in northern Jordan, namely, King Abdullah University Hospital (KAUH) and Princess Basma Hospital (PBH). The study composed of an interventional group, in which patients received a supplemented medication reconciliation service, and a retrospective control group in which patients received the usual care that may or may not include medication reconciliation. Patients in the interventional group were recruited over a 4-month period (February to May 2023) and received medication reconciliation at admission and discharge in addition to continuous medication review during their admissions (supplemented medication reconciliation). In the retrospective control group, patients were identified retrospectively using medical records. Patients who were admitted with CKD diagnosis during the years 2015–2019 were identified as those discharged with International Classification of Diseases, 10th

Revision, Clinical Modification (ICD-10-CM) Diagnosis Code of ‘N18’ (used for CKD admissions). All patients with complete data required for retrospective medication reconciliation were included in the control group. Demographic, clinical, medical and pharmacy data of each patient were retrieved from medical records at admission. Prescriptions 6-months before admission, admission medication lists, medication lists given during hospitalisation, prescriptions at discharge and patient medication prescriptions 9-months after discharge were extracted and analysed to assess discrepancies upon admission and discharge in a methodology adopted from a previous study (Volpi et al., 2021). Figure 1 illustrates the timeframe and outcomes measured in the study groups.

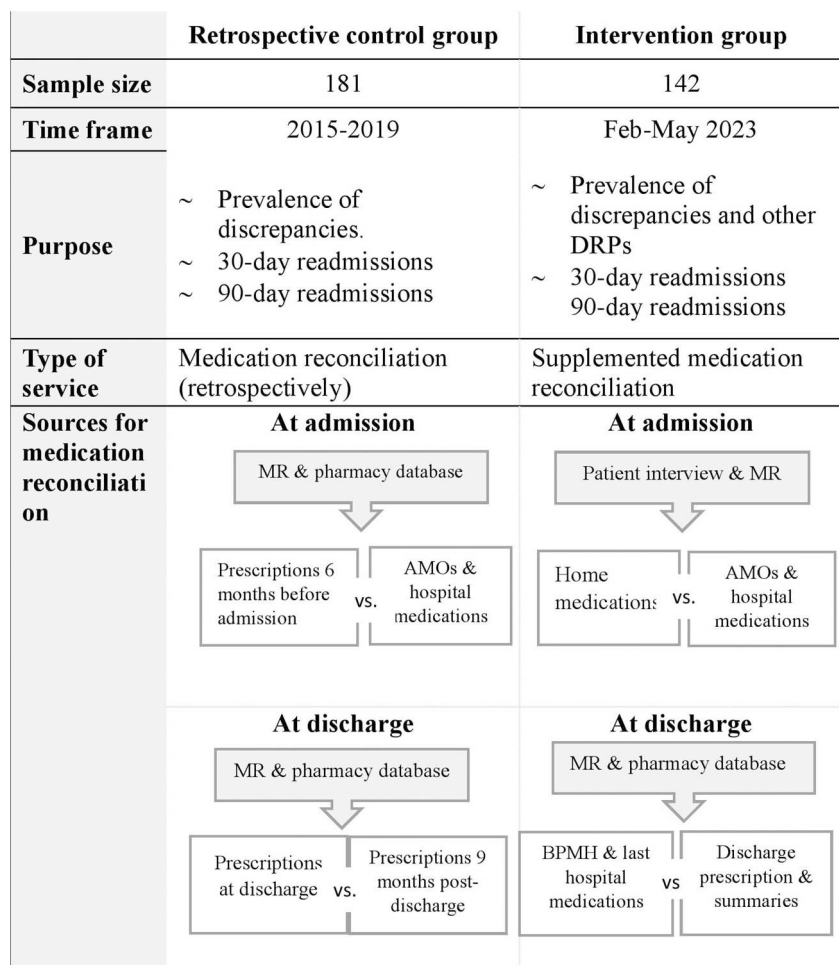


Figure 1. Summary of timeline and procedure for each group. AMOs: admission medication orders, MR: medical records, BPMH: best possible medication history, VS: versus.

Inclusion and exclusion criteria

Medical records at KAUH and PBH were reviewed to select eligible patients. The study targeted the adult (18 years or older) admitted patients with CKD (stages 2–5 including dialysis), who were using at least three chronic medicines at home, and were hospitalised for at least 48 h. Patients with kidney transplantation were excluded. Patients who have difficulties in responding to interviews and did not have family member or caregiver on behave, and those who refused to provide a written informed consent were excluded from the intervention group.

Medication reconciliation process supplemented with medication review

Patients in the intervention group received a medication reconciliation model across the transitions of care during the index hospitalisation in the internal medicine ward plus medication review for possible DRPs. This service was provided by a trained clinical pharmacist in the internal medicine wards. Eligible subjects received enough information about the purpose and details of the study before signing the informed consent form. At admission, demographic, clinical, and medical information for each enrolled patient were collected from the medical records. Patients or their caregivers were then interviewed to verify demographics, medical history, and pre-admission medications list which was further verified using all other available sources such as bottles, prescriptions, and previous medical records to generate the best possible medication history (BPMH). All comorbidities were assessed for each patient and Charlson Comorbidity Index (CCI) was calculated (Charlson et al., 1987). At discharge, the best possible discharge medication plan (BPMDB) was created from the BPMH, the last medication list during index hospitalisation, and the new medications planned to be started upon discharge. The BPMH, medication reconciliation form, and BPMDB were adapted from the CPSI/ISMP Canada (*Canadian Patient Safety Institute (CPSI) and the Institute for Safe Medications Practices Canada (ISMP Canada), Medication Reconciliation in Acute Care Getting Started Kit edition 4, 2017, 69P*), with minor modifications. The procedure of the supplemented medication reconciliation for the intervention group was divided into several steps as described in [Supplemental Table S1](#) and depicted in [Figure 2](#).

Study outcomes

The primary outcome in this study was the likelihood of all-cause 90-day readmission as extracted from medical records for both study groups. In addition, 30-day readmission was also evaluated. All assessed readmissions were unscheduled, excluding hospital-based dialysis sessions. Medication discrepancies were evaluated at admission and at discharge for patients recruited in the

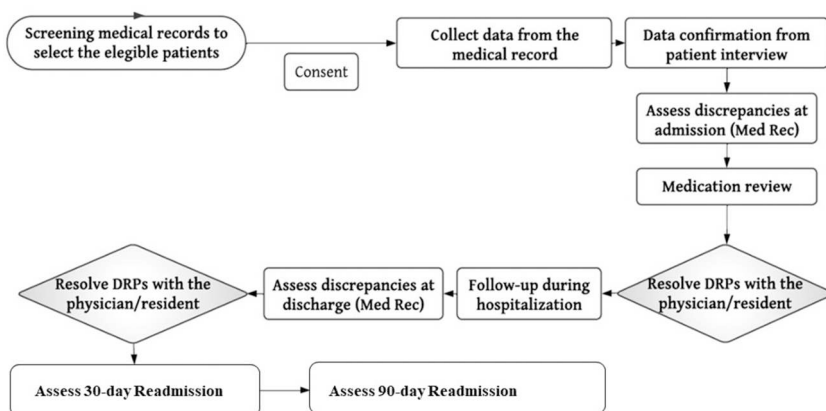


Figure 2. Supplemented medication reconciliation process flow.

intervention group (prospectively) and for patients in the retrospective control group (retrospectively). Classification of discrepancies was made using the Medication Discrepancy Taxonomy (MedTax) system (Almanasreh et al., 2020). Decisions about the response to clinical pharmacist recommendations in the intervention group based on identified discrepancies were categorised into accepted, not accepted, or potentially accepted. Potential accepted status indicated situations in which the clinical pharmacist could not communicate the intervention to the responsible physician/resident on time.

Sample size calculation

Based on a previously reported 90-day readmission proportion of 0.40 among CKD patients (Tesfaye et al., 2019b), about 264 subjects were required to detect at least 40% decrease in readmission proportion considering a power of 0.8, and a significance level of 0.05.

Statistical analysis

Descriptive statistics, such as percentages and arithmetic means with standard deviations were calculated to describe patient characteristics and outcome measures. Unadjusted comparisons between the intervention and control groups were conducted using the t-test, non-parametric (Mann-Whitney) test, or chi-square as appropriate. Compared to the retrospective control group, multi-variable linear regression and logistic regression models were conducted to evaluate the impact of medication reconciliation on reducing the number of discrepancies at discharge and on the likelihood of readmission, respectively. In all regression models, the effect of the intervention was evaluated while adjusting for age, gender, marital status, body

mass index, CCI, and renal replacement therapy (dialysis). Data were analysed using Stata version 17 software (StataCorp, 2021). *P*-values less than 0.05 were considered statistically significant.

Results

Patients' socio-demographics and clinical characteristics

In total, 323 eligible patients with CKD were included in the study, of whom 142 were recruited prospectively in the intervention period and 181 were included in the retrospective control group. Among eligible patients, 4 patients refused to participate in the intervention group with a response rate of 97.3%. The average age ($\pm$ standard deviation) of patients in the interventional group was 57.16 ± 15.96 years, while in the retrospective cohort group, it was 62.54 ± 15.75 years. Patients were distributed across CKD stages 2–5, with a bit more than 50% were on dialysis in both groups. On average, the sample had 5.4 ± 2.2 comorbid conditions. Average CCI was 5.84 ± 2.6 for the whole sample and was not significantly different between the two groups (*P* value = 0.140). Detailed demographic and clinical characteristics for the interventional and the retrospective control groups are presented in Table 1.

Table 1. Demographic characteristics of the recruited sample.

Variable	Total N = 323	Intervention N = 142	Control (Retrospective) N = 181	<i>P</i> value
Gender n (%)				0.295
Female	131 (40.6)	53 (37.3)	78(43.09)	
Male	192 (59.4)	89 (62.7)	103(56.91)	
Age, years (M $\pm$ SD)	60.2 (16.0)	57.16 (15.96)	62.54 (15.75)	0.003
BMI n (%)				0.504
<18.5	8 (2.9)	6 (4.23)	2 (1.56)	
18.5–24.9	85 (31.5)	41 (28.87)	44 (43.38)	
25–29.9	78 (28.9)	42 (29.58)	36 (28.13)	
> 29.9	99 (36.7)	53 (37.32)	46 (35.94)	
Marital status				<0.001
Married	265 (87.2)	109 (76.76)	156 (96.3)	
Not married	39 (12.8)	33 (23.24)	6 (3.7)	
Reason for admission				0.068
AKI on Top of CKD related problems	127 (39.3)	42 (29.58)	85 (46.9)	
CKD/dialysis related problems	108 (33.4)	71 (50.0)	37 (20.4)	
Others	88 (27.2)	29 (20.42)	59 (32.6)	
CKD stage				0.172
Stage 2	3 (0.93)	2 (1.41)	1 (0.55)	
Stage 3a	13 (4.02)	3 (2.11)	10 (5.52)	
Stage 3b	31 (9.6)	12 (8.45)	19 (10.5)	
Stage 4	85 (26.3)	32 (22.54)	53 (29.28)	
Stage 5	191 (59.1)	93 (65.5)	98 (54.14)	
CCI (M $\pm$ SD)	5.84 ± 2.6	6.08 ± 2.93	5.65 ± 2.2	0.140
LOS (M $\pm$ SD)	9.26 ± 8.8	8.9 ± 8.5	9.5 ± 8.8	0.557

AKI: acute kidney injury, CKD: chronic kidney disease, CCI: Charlson comorbidity index, LOS: length of stay.

The prevalence of discrepancies in CKD patients

In the interventional group, a total of 358 medication discrepancies were identified based on the medication reconciliation at admission and at discharge. The average number of discrepancies was 2.5 ± 2.2 per admission. The most common types of discrepancies were drug omission (70.7%), followed by discrepancies in strength/total daily dose/frequency (14.8%) and unnecessary drug addiction (6.4%). Almost half of the interventions were considered as potential to be accepted but were not communicated for physicians (50.5%). Around 38.4% of recommendations were accepted and implemented by physicians/residents.

In the retrospective cohort control group, a total of 471 discrepancies were identified, and the average number of discrepancies was 2.6 ± 2.5 per admission. Drug omission, followed by discrepancies in strength/total daily dose/frequency and therapeutic class substitution were the most common types of discrepancies. [Supplemental Fig. S1](#) depicts discrepancies distribution according to MedTax classification for both the intervention and the retrospective control groups.

Impact of medication reconciliation intervention on decreasing discrepancy numbers at discharge

The average decrease in discrepancy numbers between admission and discharge in the intervention group was 0.73 compared to -0.91 in the retrospective control group, $P < 0.001$. Adjusting for potential confounders, the decrease in discrepancy numbers between admission and discharge was higher in the intervention group by 1.66 discrepancies compared to the retrospective control group ([Table 2](#)).

Impact of medication reconciliation intervention on hospital readmission

Compared to the retrospective control group, the interventional group, had significantly lower odds of all-cause 90-day readmissions (OR = 0.41, $P = 0.002$), adjusting for age, gender, CCI, body mass index (BMI), marital status, and receiving dialysis. However, no significant impact was seen on 30-day readmission as shown in [Table 3](#).

Discussion

The current study showed a positive impact of the medication reconciliation supplemented with medication review on CKD patients' clinical outcomes. Supplemented medication reconciliation was effective in reducing the

Table 2. Predictors of the decrease in discrepancy numbers between admission and discharge.

Decrease in discrepancies	Coefficient	95% CI		P value
Intervention group	1.66	1.07	2.26	<0.001
Age	−0.01	−0.03	0.02	0.554
Male gender	0.02	−0.56	0.61	0.936
CCI	0.07	−0.09	0.22	0.411
Dialysis	0.32	−0.24	0.87	0.263
BMI				
Overweight	0.11	−0.60	0.83	0.760
Obese	0.29	−0.36	0.94	0.384
Married	0.28	−0.64	1.20	0.548

Analysis was conducted after excluding patients for whom medication reconciliation was not conducted at discharge (49 among the intervention group and 41 among the retrospective cohort group).

BMI: body mass index; CCI: Charlson comorbidity index.

Table 3. Predictors of all-cause readmissions within one month and three months after discharge (adjusted analysis).

All cause readmission	Odds ratio	95% CI		P value
30-day readmission				
Intervention group	0.62	0.32	1.19	0.150
Age	1.00	0.97	1.02	0.866
Male gender	0.74	0.39	1.39	0.352
CCI	1.14	0.97	1.34	0.104
Dialysis	1.05	0.57	1.95	0.867
BMI				
Overweight	1.26	0.62	2.58	0.524
Obese	0.65	0.31	1.34	0.240
Married	0.80	0.31	2.09	0.648
90-day readmission				
Intervention group	0.41	0.23	0.72	0.002
Age	1.00	0.98	1.02	0.902
Male gender	0.69	0.40	1.19	0.180
CCI	1.03	0.89	1.19	0.699
Dialysis	1.18	0.69	2.01	0.547
BMI				
Overweight	1.06	0.56	2.02	0.862
Obese	0.71	0.38	1.32	0.279
Married	1.21	0.52	2.86	0.657

Patients who died during the index admission were excluded (13 patients; 5 from the intervention group and 8 from the retrospective cohort group).

BMI: body mass index; CCI: Charlson comorbidity index.

number of medication discrepancies at discharge. The majority of identified discrepancies were due to drug omission at admission and discrepancies in strength, total daily dose, or frequency. A positive impact of medication reconciliation supplemented with medication review was observed on patients' readmission within 3 months of discharge.

The current study demonstrated that the most frequent type of discrepancies was drug omission, followed by discrepancy in strength/total daily dose/frequency. This was in agreement with other studies regarding the frequency of drug omission among non-dialysis CKD patients (Hawley et al., 2019), and in

the general CKD population (Song et al., 2021). Chen et al found that the highest proportion of discrepancies among haemodialysis patients was associated with incorrect dose/frequency (42%), followed by commissions (38%) and omissions (21%) (Chan et al., 2015). Most discrepancies occurred at admission, a finding consistent with previous reports indicating that many discrepancies were due to incomplete recording of medication histories at admission (Tam et al., 2005). Additionally, addressing admission discrepancies and DRPs during the hospital stay would likely have a positive impact at discharge. These findings emphasise the importance of conducting systematic medication reconciliation upon admission, especially for high-risk groups such as CKD patients.

The acceptance rate of interventions observed in the current study (38.4%) was relatively comparable or lower than the rate reported previously among studies conducted in Jordan which reported 38% (Hammour et al., 2022), 54.7% (Salameh et al., 2019), 69.2% (AbuRuz et al., 2013), and 83.9% (Tahaineh & Khasawneh, 2018) in various populations with different outcomes. The relatively low acceptance rate was related in part to the limited timely access to discharge summaries and prescriptions as well as the difficulty in reaching the responsible physician in some cases. Consequently, about half of the DRPs were labelled as potential to be accepted in the current study.

Supplemented medication reconciliation was effective in reducing the number of discrepancies at discharge compared to the CKD sample which received the usual care. Resolving discrepancies at admission and implementing a continuous medication review during the admission played a potentially important role in reducing discrepancies from admission to discharge. Though, a previous randomised clinical trial in Jordan evaluated the impact of medication reconciliation on decreasing medication discrepancies after discharge from the surgical ward, and showed a reduction in discrepancy number in both the intervention group and the control group (Hammour et al., 2022). The current study focused specifically on CKD patients, and revealed that the intervention group experienced a significantly greater reduction in medication discrepancies. This finding further underscores the clinical value of this service in the CKD context. It is well-documented in the literature that patients with CKD have significantly higher readmission rates compared to the general populations (Mathew et al., 2015; USRDS, 2021). This study showed a statistically significant difference in all-cause readmissions between the intervention and control groups at 3 months after discharge. This result is strongly supported by the literature especially in those with reasonable follow-up duration (Chia et al., 2017; Pai et al., 2009). For instance, Chi et al conducted a comprehensive medication review with pharmacist involvement for 134 haemodialysis patients in a nephrology outpatient clinic (Chia et al., 2017), and reported reduced unplanned admissions within a 6-months follow up period. Also, Pai et al delivered regular medication review to 57 haemodialysis patients up to 2 years and reported similar results (Pai et al., 2009). Additional

prospective study involved 50,000 patients undergoing haemodialysis found a significant reduction in rates of readmissions by 0.93 (95% CI, 0.90–0.96) after 30 months of follow-up (Weinhandl et al., 2013). Nevertheless, other published articles on CKD patients showed negligible effect on all-cause readmissions (Jasinska-Stroschein, 2022; Song et al., 2021; Tuttle et al., 2018). This may be justified partially by the variability in sample sizes, different interventions provided, and/or variable follow-up periods.

Given the positive outcomes associated with medication reconciliation, its global implementation – particularly for high-risk patients such as those with CKD or polypharmacy – is essential for improving patient care. However, despite its well-established benefits, global implementation faces significant challenges due to disparities in healthcare infrastructure, policies, and resource availability (Katoue & Ker, 2018; Redmond et al., 2020; van Sluisveld et al., 2012). Barriers to the standardised adoption of medication reconciliation worldwide include limited resources in low-income settings, unequal access to trained pharmacists, insufficient formal training for healthcare professionals, workforce shortages, health literacy challenges, and inconsistent medication reconciliation policies across countries (Katoue & Ker, 2018; Redmond et al., 2020; van Sluisveld et al., 2012).

The current study adds to the little existing international literature that evaluated the impact of medication reconciliation in CKD patients. Whereas most of these studies focused on end-stage renal disease, the impact of this service is still largely understudied in other CKD stages (Al Raiisi et al., 2019). This study was the first in employing the Quasi experimental design using a retrospective control group in which discrepancies were evaluated and compared with the interventional group without any ethical concerns or contamination. In addition, this study was the first in examining the impact of medication reconciliation on relevant clinical outcomes in admitted patients distributed across various CKD stages. Despite the rigorous efforts, this study has several limitations. The current controlled trial was not randomised which might introduce the risk of selection bias. However, multivariable regression models were applied to control potential confounders. Limited timely access was the main restrictor for the clinical pharmacist. This was related to alternating the days between the two hospitals, and consequently decreasing the availability of the pharmacist at the time of discharge for some patients. In the real-life setting, having a full-time clinical pharmacist is expected to display an effective role and profound effects on patients' outcomes. In the current study, all-cause readmissions were evaluated without assessing the relation to DRPs. However, we believe that resolving DRPs including discrepancies during hospitalisation can have an insightful impact on the general health status, for which all-cause readmission is a valid indicator. For the retrospective control group, we assumed that in some cases, patients may have purchased medications outside the hospital,

making those records unavailable in the retrieved pharmacy records. Similarly, readmissions to other institutions could not be verified retrospectively. However, it is highly likely that most patients would be readmitted to the same institution due to their insurance coverage and geographic proximity. Finally, including hospitals in the north of Jordan limits the generalisability of study findings to health practices in the south and middle of Jordan, though not expected to differ widely.

Future efforts should prioritise expanding this integrated service across diverse healthcare settings and assessing its long-term impact on clinical, humanistic, and economic outcomes to enhance its global applicability. Additionally, promoting multidisciplinary collaboration and integrating medication reconciliation into routine pharmacy practice is essential to ensuring the safe and effective management of CKD and other chronic diseases. Further studies may help clarify the causal relationship between these discrepancies and DRPs, as well as their link to cause-specific readmissions and mortality. Furthermore, additional studies are needed to evaluate a wider scope of pharmacist-led interventions, including bedside counselling at discharge, telephone follow-ups, patient education, and providing take-home medication lists, with a particular focus on CKD patients.

Conclusion

In conclusion, this study demonstrated that CKD inpatients are very susceptible to DRPs. The current findings suggest that medication reconciliation with medication review service can reduce discrepancy numbers at discharge as well as readmission rates in CKD patients. Multidisciplinary healthcare team that integrates clinical pharmacist is highly recommended for better health outcomes among inpatients with CKD.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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